Question 9: Create a Pivot Table and Find Total Sale Amount Region-wise, Manager-wise, Salesman-wise

Aim

To create a Pivot Table using Pandas and calculate the total sale amount based on Region, Manager, and Salesman.

Algorithm

1. Import the Pandas library.
2. Read the dataset9.txt file.
3. Store the data in a DataFrame.
4. Create a Pivot Table using pd.pivot_table().
5. Set **Sale_Amount** as the values column.
6. Set **Region, Manager, and Salesman** as the index.
7. Use the **sum** aggregation function.
8. Display the Pivot Table.

Code:

```
import pandas as pd
# Read the dataset
df = pd.read_csv(r"C:\Users\HP\Downloads\dataset9.txt", sep="\t")

# Create Pivot Table
pivot = pd.pivot_table(
    df,
    values="Sale_Amount",
    index=["Region", "Manager", "Salesman"],
    aggfunc="sum"
)
print(pivot)
```

Output:

```
...
                Ezhil        35000
                Isha         32000
South   Priya   Bala         30000
                Fathima      27000
                John         29000
West    Meena   Divya        28000
                Hari         26000
```

Result

Thus, the Pivot Table was successfully created, and the total sale amount was displayed region-wise, manager-wise, and salesman-wise.

Question 10: Highlight Negative Numbers Red and Positive Numbers Black

Aim

To create a DataFrame containing random values and highlight negative numbers in **red** and positive numbers in **black** using Pandas Styling.

Algorithm

1. Import the Pandas and NumPy libraries.
2. Generate a DataFrame of 10 rows and 4 columns with random integers.
3. Define a function to assign red color to negative values and black color to positive values.
4. Apply the function using style.map().
5. Display the styled DataFrame.

```
C: > Users > HP > Downloads > boxplot.py > ...
import pandas as pd
import numpy as np
# Generate random values
np.random.seed(10)
df = pd.DataFrame(
    np.random.randint(-50, 51, size=(10,4)),
    columns=["A","B","C","D"]
)
# Function to highlight values
def color_negative(val):
    if val < 0:
        return "color:red"
    else:
        return "color:black"

print(df)
# Display styled dataframe
styled_df = df.style.map(color_negative)
styled_df
```

Output:

```
...
3 -14 -34  50 -39
4   4  38  12 -17
5  22  50  28  -1
6   1   4  27  19
7 -37 -25 -37  42
8  36 -20 -20  39
9 -38  15 -19   7
```

Result

Thus, a DataFrame with random values was created successfully, and negative numbers were highlighted in **red** while positive numbers were highlighted in **black** using Pandas styling.